never see particles, but such states will have more energy overall than the vacuum state.

You might be tempted to ask: But are the particles *really there*? How can there be zero particles in the universe as a whole, and yet we might see particles when we look in any particular location?

But we're not dealing with a theory of particles; it's a theory of fields. Particles are what we see when we observe the theory in particular ways. We shouldn't be asking, "How many particles are there, really?" We should be asking, "What are the possible measurement outcomes when we observe a quantum state in this specific way?" A measurement of the form "How many particles are there in the entire universe?" is fundamentally different from one of the form "How many particles are there in this room?" So different that, just as for position and momentum, no quantum state will give definite answers for both questions at the same time. The number of particles we see isn't an absolute reality, it depends on how we look at the state.

This leads us directly to an important property of quantum field theory: the entanglement between parts of the field in different regions of space.

Imagine dividing the universe into two regions by drawing an imaginary plane somewhere in space. Call the regions "left" and "right" for convenience. Classically, since fields live everywhere, to construct any particular field configuration we would have to specify what the field is doing both in the left region and in the right region. If there is a mismatch of the value of the fields across the boundary, that will correspond to a sharp discontinuity in the profile of the field overall. That's conceivable, but it costs energy for the field to change from point to point, so a discontinuous jump implies a large amount of energy at that point. This is why ordinary field configurations tend to vary smoothly, rather than suddenly.

At the quantum level, the classical statement "The field value tends to match across the boundary" turns into "The fields in the left and right regions tend to be highly entangled with each other." We can consider